

Numerical Analysis

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Part I

Algorithms

Chapter 1

Introduction

Chapter 2

Solutions of Equations in One Variable

2.1 Bisection Method

2.1.1 Main Idea

Bisection method is based on IVT, Theorem 4.2.2.1. Given a function f , we start with initial interval $[a, b]$ such that $f(a)f(b) < 0$, checking the sign of intermediate value $f(\frac{a+b}{2})$. If the sign equals that of $f(a)$, we set $a = \frac{a+b}{2}$, else we set $b = \frac{a+b}{2}$.

2.1.2 Pseudocode

Algorithm 1: Bisection Method

Input : $f(\cdot)$ a function which we seek for its root
 $a < b \in \mathbb{R}$ endpoints, satisfying $f(a)f(b) < 0$
 N number of iterations
 TOL tolerance

Output: p approximate solution

```
1 for  $i = 0, 1, 2, \dots, N - 1$  do
2    $p \leftarrow \frac{a+b}{2}$ 
3   if  $f(p) = 0$  or  $\frac{b-a}{2} < TOL$  then
4     return  $p$  // procedure completed succesfully
5   else if  $f(a)f(p) > 0$  then
6      $a \leftarrow p$ 
7   else
8      $b \leftarrow p$ 
9 return failure after  $N$  iterations
```

2.1.3 Implementation Details

2.1.3.1 Stopping Procedure

One can change the stopping procedure of algorithm 1. With tolerance $\varepsilon > 0$, followings are examples of other stopping procedure:

$$|p_N - p_{N-1}| < \varepsilon \quad (2.1)$$

$$\frac{|p_N - p_{N-1}|}{|p_N|} < \varepsilon, \quad p_N \neq 0 \quad (2.2)$$

$$|f(p_N)| < \varepsilon \quad (2.3)$$

But, there are some difficulties. There are sequences $\{p_n\}$ where $|p_n - p_{n-1}|$ converges to zero while $p_n \rightarrow \infty$. It is also possible for $f(p_n)$ to be close to zero while p_n differs significantly from p . The second procedure is the best when we know nothing about f and p .

2.1.3.2 Overflow and Underflow

It is recommended to use following computation for calculating $\frac{a+b}{2}$:

$$p = a + \frac{b - a}{2}. \quad (2.4)$$

When $b - a$ is near maximum precision, a round off error is still in presense, but it would not affect the value of p so much. However, when $a + b$ is near maximum precision, $\frac{a+b}{2}$ might not even produce a result in interval $[a, b]$.

Using **sign** function for determining the equality of sign can prevent overflow or underflow in the multiplication of $f(a)$ and $f(p)$.

2.1.4 Pros and Cons

Pros:

- Very easy to implement;

Cons:

- Slow convergence rate. It has linear convergence rate;
- Cannot extend to multi-dimensional functions;
- Does not exploit additional informations such as derivatives. This leads to slow convergence rates.

2.1.5 Convergence

TODO: Bisection Method Convergence Proof

2.2 Fixed-Point Iteration

2.2.1 Main Idea

Fixed-point iteration is based on Theorem 2.2.5.1. The definition of fixed point is as following:

Definition 2.2.1.1 (Fixed Point)

Let $X, Y \subset \mathbb{R}$. Then the **fixed point** of a function $f : X \rightarrow Y$ is a point $p \in X$ such that

$$f(p) = p. \quad (2.5)$$

There are sufficient conditions for the existence and uniqueness of a fixed point.

Theorem 2.2.1.1 (Existence and Uniqueness of a Fixed Point)

Let $X, Y \subset \mathbb{R}$. Let $a < b \in X$, and let $g : X \rightarrow Y$ be a function. Then, following holds:

1. If $g \in C[a, b]$ and $g(x) \in [a, b]$ for all $x \in [a, b]$, then g has at least one fixed point in $[a, b]$.
2. If, in addition, $g'(x)$ exists on (a, b) and a constant $0 < k < 1$ exists with $|g'(x)| \leq k$ for all $x \in (a, b)$, then there exists a unique fixed point in (a, b) .

Proof. Can be proved with IVT and MVT. □

One may wonder what fixed point does with the root of a function. By modifying the equation $f(x) = 0$ a bit, we can reformulate it into an equation about fixed point, for example, $f(x) + x = x$ or $x - 3f(x) = x$.

TODO: Figures of fixed point iteration

2.2.2 Pseudocode

Algorithm 2: Fixed-Point Iteration

Input : $g(\cdot)$ a function which we seek for its fixed point
 p_0 initial approximation
 N number of iterations
 TOL tolerance

Output: p approximate solution

// initialize

```
1  $p \leftarrow p_0$ 
2 for  $i = 0, 1, 2, \dots, N$  do
3    $p \leftarrow g(p)$ 
4   if  $|p - p_0| < TOL$  then
5     return  $p$  // procedure completed successfully
6 return failure after  $N$  iterations
```

2.2.3 Implementation Details

2.2.4 Pros and Cons

Pros:

- Easy to implement;

Cons:

- Slow convergence rate. It has linear convergence rate. Methods of increasing convergence rate is treated in ??.
- Depends on the modified equation. Convergence result depends on the modified equation of $f(x) = 0$;

2.2.5 Convergence

2.2.5.1 Fixed-Point Theorem

Theorem 2.2.5.1 (Fixed-Point Theorem)

Part II

Theories

Chapter 3

Introduction

Chapter 4

Calculus

4.1 Introduction

This chapter reviews calculus briefly.

4.2 Limits and Continuity

4.2.1 Limit

Definition 4.2.1.1 (Limit)

Let $X \subset \mathbb{R}^n, Y \subset \mathbb{R}^m$. Then, the function $f : X \rightarrow Y$ is said to have the **limit** L at x_0 if and only if

$$\text{for all } \varepsilon > 0, \text{ there exists } \delta > 0 \text{ such that } \|f(x) - L\| < \varepsilon \Rightarrow \|x - x_0\| < \delta \quad (4.1)$$

4.2.2 Continuity

Definition 4.2.2.1 (Continuity)

Let $X \subset \mathbb{R}^n, Y \subset \mathbb{R}^m$. Let $A \subset X$. Then, the function $f : X \rightarrow Y$ is said to be **continuous** on set A if and only if

$$\text{for all } a \in A, \lim_{x \rightarrow a} f(x) = f(a), \quad (4.2)$$

and denote it as $f \in C^0(X)$ where $C^0(X)$ is a set of continuous functions on X .

Theorem 4.2.2.1 (Intermediate Value Theorem)

Let $X \subset \mathbb{R}, Y \subset \mathbb{R}$. Let $a < b \in \mathbb{X}$. Let $f : X \rightarrow Y$ be a continuous function on $[a, b]$. Then, if $K \in \mathbb{R}$ is any number between $f(a)$ and $f(b)$, then there exists a number $c \in [a, b]$ such that $f(c) = K$.